

Write your questions
and thoughts here!

Squeeze Theorem: a.k.a. “Sandwich Theorem” or “Pinching Theorem”

$$\text{If } g(x) \leq f(x) \leq h(x)$$

$$\text{and if } \lim_{x \rightarrow a} g(x) = L \text{ and } \lim_{x \rightarrow a} h(x) = L$$

$$\text{then } \lim_{x \rightarrow a} f(x) = L$$

1. Find $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x^2}\right)$

2. Let g and h be the functions defined by $g(x) = -x^2 + 2x - 3$ and $h(x) = 2x + 1$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow 2} f(x)$?

3. Let g and h be the functions defined by $g(x) = \cos\left(\frac{\pi}{2}x\right) + 2$ and $h(x) = x^2 + 3$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-1 \leq x \leq 5$, what is $\lim_{x \rightarrow 0} f(x)$?

Let f , g , and h be the functions defined by $f(x) = \frac{1 - \cos x}{2x}$, $g(x) = x^3 \sin\left(\frac{1}{x}\right)$, and $h(x) = \frac{x}{\sin x}$ for $x \neq 0$. All of the following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

4. $-\frac{1}{2} \leq f(x) \leq x^2 + \frac{1}{2}$

5. $-x^3 \leq g(x) \leq x^3$

6. $1 - x^2 \leq h(x) \leq x^2 + 1$

1.8 The Squeeze Theorem

Calculus

Practice

Evaluate each limit.

1. $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right)$

2. $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x^2}\right)$

3. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x^2}\right)$

4. Let g and h be the functions defined by $g(x) = x^2 - 3x$ and $h(x) = 2 - 2x$.
If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow 2} f(x)$?

5. Let g and h be the functions defined by $g(x) = \cos(\pi(x + 2)) - 3$ and $h(x) = \frac{x^2}{2} + x - \frac{7}{2}$.
If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-2 \leq x \leq 0$, what is $\lim_{x \rightarrow -1} f(x)$?

6. Let g and h be the functions defined by $g(x) = x^2 + x - 1$ and $h(x) = -x^2 - 4x - 2$.
If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow -1} f(x)$?

7. Let g and h be the functions defined by $g(x) = -x^2 - 2x + 5$ and $h(x) = 2x^2 - x - 4$.
If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow -2} f(x)$?

8. Let g and h be the functions defined by $g(x) = \sin\left(\frac{\pi}{2}(x + 1)\right) - 1$ and $h(x) = \cos(\pi x) - 3$.
If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow 2} f(x)$?

9. Let g and h be the functions defined by $g(x) = x^2$ and $h(x) = \cos(x)$.
If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-0.4 \leq x \leq 0.4$, what is $\lim_{x \rightarrow 0} f(x)$?

10. Let f and g be the functions defined by $f(x) = \frac{\sin x}{5x}$ and $g(x) = x^2 \cos\left(\frac{1}{x^3}\right)$ for $x \neq 0$. The following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

a. $5(\sin(\pi(x+1))) \leq f(x) \leq \frac{1}{5}$

b. $-x^2 \leq g(x) \leq x^2$

11. Let f and g be the functions defined by $f(x) = \frac{\cos x - 1}{x^2}$ and $g(x) = x^2 \sin\left(\frac{1}{x}\right)$ for $x \neq 0$. The following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

a. $\frac{1}{5} \leq f(x) \leq \frac{1}{2}$

b. $-x^2 - \frac{1}{2} \leq f(x) \leq x^2 - \frac{1}{2}$

c. $-x^2 \leq g(x) \leq x^2$

d. $-\frac{1}{x} \leq g(x) \leq \frac{1}{x}$

12. Let f and g be the functions defined by $f(x) = \frac{x^2 \sin x}{x}$ and $g(x) = x \cos\left(\frac{1}{|x|}\right)$ for $x \neq 0$. The following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

a. $-x \leq f(x) \leq x$

b. $-\frac{1}{x} \leq g(x) \leq \frac{1}{x}$

c. $2^{-x} - 1 \leq g(x) \leq 2^x - 1$